This Zenodo deposit contains a publicly available description of the Dataset:

"Bulk RNAseq of mouse substantia nigra of wild type and PINK1 KO mice after C.rodentium infection".

Dataset Description:

We performed bulk RNAsequencing of substantia nigra from wild type and PINK1 KO mice 26-days post C.rodentium infection. This dataset contains raw fastq files from striatal cells, sorted into 4 groups namely wild type and PINK1 KO uninfected and infected mice. Sequencing was performed using NextSeq500. FASTQs generated from sequencing output were aligned to the mm10 reference genome using STAR aligner.

Authors:

- Stratton, Jo Anne; ORCID:0000-0002-1205-1353; Montreal Neurological Institute-Hospital, McGill University
- Mukherjee, Sriparna; ORCID:0000-0002-1299-4343; University of Montreal
- Trudeau, Louis-Eric; ORCID:0000-0003-4684-1377; University of Montreal

ASAP Team: Desjardins

Dataset Name: desjardins-mouse-bulk-rnaseq-nigra-pink1, v1.0

Principal Investigator: Louis-Eric Trudeau louis-eric.trudeau@umontreal.ca

Dataset Submitter: Moein Yaqubi moein.yaqubi@mcgill.ca

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ASAP Project: A single Citrobacter rodentium infection in PINK1 knockout and wild type mice leads to regional blood-brain-barrier perturbation and glial activation without dopamine neuron axon terminal loss

Project Description: A growing body of research supports the hypothesis of links between immune system activation and the development of Parkinson's disease (PD). A recent study revealed that repeated gastrointestinal infection with Citrobacter rodentium can lead to PD-like motor dysfunction in Pink1 knockout (KO) mice and immune cell entry in the brain. With the objective of better understanding the mechanisms leading to immune attack of the brain in this model, we evaluated the hypothesis that such mild infections are sufficient to increase blood brain barrier (BBB) permeability and cause brain inflammation. Pink1 wild-type (WT) and KO mice were infected with Citrobacter rodentium and at day 13 and 26 post infection, we conducted gadolinium-enhanced magnetic resonance imaging (MRI) to identify signs of BBB permeability changes. Quantification of MRI results provided evidence of increased blood-brain barrier permeability in both WT and Pink1 KO mice at 26 days after the infection in the striatum, dentate gyrus, somatosensory cortex, and thalamus. This was not accompanied by any change in global expression of tight-junction proteins or in markers of the integrity of the dopamine (DA) system in the striatum at both time points. However, chronic microglial activation was detected at day 26 post infection, accompanied by an elevation of the inflammatory mediators eotaxin, IFNγ, CXCL9, IL-17 and MIP-2 in the striatum, accompanied by an elevation of IL-17 and CXCL1 in the serum of Pink1 KO mice. Neutrophil infiltration in the brain of infected mice was also noted at day 26 post infection, as revealed by immune cell profiling by flow cytometry. Finally, a bulk RNA-seq transcriptome analysis revealed that gene sets related to synaptic function were particularly influenced by the infection and that inflammation-related genes were upregulated by the infection in the Pink1 KO mice. Our results support the hypothesis that even after mild gastro-intestinal infection, increased BBB permeability could contribute to perturbations of brain homeostasis including altered expression of synaptic genes, increased microglial activation and the establishment of a chronic state of brain inflammation. Such perturbations could potentially act as a first hit for subsequent induction of PD pathology in the context of aging in genetically susceptible individuals.

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